

Intent and sentiment aware information retrieval

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Abstract

This work addresses the challenge of supporting non-expert users in understanding financial questions by designing an intent-aware and sentiment-diverse information retrieval system. The goal is to mitigate the risks of misinterpretation, emotional bias, and one-sided advice commonly found in financial forums. We propose a hybrid retrieval pipeline that combines traditional lexical models (BM25) with neural re-ranking, semantic query expansion, and document augmentation. For subjective, advice-seeking queries, we introduce intent classification and sentiment-aware diversity re-ranking to expose users to contrasting viewpoints, while objective queries prioritize factual accuracy. Experiments are conducted on a financial Community Question Answering dataset of approximately 57,000 documents and 1,200 queries, evaluated using standard IR metrics such as MAP, Precision, Recall, and nDCG. Results show that neural re-ranking significantly improves relevance, while sentiment-diverse retrieval slightly reduces traditional metrics but enhances result diversity.

1 Introduction and Motivation

In this project, we address the problem of providing non-expert users with effective tools to better understand financial-specific documents, with the ultimate goal of supporting more informed decisions and considerations. The financial domain is high-risk, since conflicting emotional factors and rational analysis can lead to poor or even catastrophic outcomes - not only for non-experts, but sometimes even for professionals.

To develop and evaluate such tools, we rely on a collection of domain-specific questions and answers obtained from a financial Q&A website. This dataset allows us to establish a baseline for information retrieval performance and to systematically test our proposed ideas. Our intended users are non-expert individuals, as they are particularly vulnerable when making financial decisions: they may conduct insufficient research, misinterpret complex information, and be influenced by the biases of other users, ultimately leading to unfavorable outcomes.

The main goal of our system is to distinguish between objective and subjective queries and documents. For objective queries, the system aims to present factual information, while for subjective queries, the system retrieves opinion-based content that reflects different viewpoints and sentiments in order to avoid presenting only one-sided, and instead support more balanced and informed decision-making

This leads to our central research question:

Does intent-aware and sentiment-diverse retrieval improve ranked document diversity and retrieval quality for financial questions compared to standard relevance-based retrieval?

To answer this question, we conduct an ablation study, allowing us to isolate the contribution of individual system components and better understand which design choices lead to measurable improvements and which do not.

2 Task and Dataset Description

The task addresses a hybrid financial question answering and information retrieval problem that necessitates distinguishing between two distinct retrieval modes: factoid retrieval and opinion-based search.

- **Factoid Retrieval:** For objective queries (e.g., "Definitions of leverage and of leverage factor" qid: 10809), the system acts as a high-precision retrieval engine, prioritizing neutral, factually accurate documents containing specific numeric or regulatory data.
- **Opinion-Based/Advisory Retrieval:** For subjective queries (e.g., "Where should I park my rainy-day / emergency fund?" qid: 4641), the problem shifts to entity-focused opinion retrieval. The goal is not to find a single "correct" answer but to retrieve a diverse set of viewpoints. This requires analyzing document stance to mitigate the risk of financial echo chambers, ensuring non-expert users are exposed to conflicting perspectives rather than potentially harmful consensus.

The experimental corpus consists of approximately 57000 documents sourced from a financial Community Question Answering platform *StockExchange*. The data is conversational in nature, representing a mix of user-generated inquiries and responses.

- **Sources:** User-generated threads discussing personal finance, market analysis, and economic policy.
- **Granularity:** The dataset exhibits extreme variance in document length, ranging from short, noise-heavy responses (2–20 tokens) to extensive conversational logs (up to 3,000 tokens). This variability reflects the "conversational" nature of support forums where critical information is often buried within long-form discussions or not given directly but perhaps in a more subtle way.
- **Content:** The text blends informal conversational markers with dense, domain-specific financial jargon.

Table 2 Query statistics

Description	Value
#Docs	57638
Docs avg length	132.90
MAX length of documents	2973.00
MIN length of documents	0
#Docs shorter than 5 words in the whole collection	73
#Short documents relevant to some queries	2
#Docs with emojis/non latin alphabet characters	49
#Docs with non standard money quantities (e.g 5k)	3747
#Docs containing mentions or hashtags	1834
#Docs with links	5946
#Docs with extended (>5 words) quoted text	14386

Description	Value
#Queries	1296
Query avg length	10.94
MAX length of queries	31
MIN length of queries	2
#Queries shorter than 5 words in the whole collection	29
#Queries with more than 1 relevant doc	428
Mean number of queries relevance assessment	2.63
#Queries with a number inside	73
#Queries with a sentiment word	111

Table 1 Document statistics

We identified three specific challenges intrinsic to this financial dataset that guided our methodology:

1. **Domain-Specific Vocabulary and Acronyms:** Financial documents rely heavily on precise identifiers (e.g., 'CREF', 'FX', 'ROE'). Standard embeddings often struggle with this specialized vocabulary, leading to **vocabulary mismatch**.
2. **The "Relevance-Diversity" Trade-off:** A unique challenge in the financial domain is that "neutrality" often signals relevance for factual queries, whereas "high sentiment" often signals speculation or noise. However, for subjective queries, ignoring sentiment results in a lack of diversity. The system must effectively toggle between "neutrality-as-relevance" (for facts) and "sentiment-as-diversity" (for opinions) to avoid promoting harmful financial speculation while still offering diverse advice.
3. **Variable Document Length:** The presence of documents up to 3,000 tokens poses a significant technical challenge for standard neural models (e.g., BERT/MiniLM), which are typically limited to 512 tokens. Naive truncation would result in the loss of critical information located at the end of long threads.

3 Methodology

3.1 Baseline Systems (Phase I)

In the first phase of our study, we focused on establishing a set of baseline systems in order to assess the effectiveness of standard information retrieval models in the context of financial question answering. To this end, we conducted three preliminary experiments aimed at analyzing the behavior of widely used ranking methods under different conditions.

In the first experiment, we evaluated the performance of BM25, PL2, and TF-IDF without applying any domain-specific constraints related to the financial sector. The goal was to assess the capabilities of these canonical retrieval models. Additionally, we experimented with RM3 combined with each model to evaluate whether query expansion through relevance feedback could lead to performance improvements.

In the second experiment, we tackled limited lexical overlap between queries and relevant documents, which harms models based on exact term matching. Because co-occurrence-based query expansion was unlikely to help, we used semantic expansion instead. Queries were expanded with synonyms from WordNet via NLTK, acronym expansions derived from the document collection, and morphological variants to improve recall. Finally, we further augmented queries using a Gemma language model to generate additional related user questions.

In the third experiment, we considered the effect of document length on retrieval performance. We observed that a subset of documents was significantly longer than the majority of the collection, potentially biasing the ranking process. To mitigate this issue, we segmented longer documents into smaller passages, with the hypothesis that passage-level retrieval would allow the system to focus on more relevant content and improve overall effectiveness.

3.2 Advanced Systems (Phase II)

In the second phase of our study, we conducted two advanced experiments aimed at improving the performance of our information retrieval system beyond the baseline models.

In the first experiment, we enhanced the preprocessing and ranking pipeline by removing hyperlinks and emojis from queries and expanding acronyms in both queries and documents. In addition, we applied a doc2query expansion technique. Using the BM25-ranked document list as our baseline, we randomly selected 20 documents per query and generated synthetic questions representing potential user queries for which those documents would be relevant. These generated questions were appended to the original query to form an expanded representation. Finally, we applied a cross-encoder model, which leverages deeper semantic understanding, to re-rank the retrieved documents and improve the final ranking quality.

In the second experiment, we kept the same preprocessing steps and the same first-stage retrieval based on BM25 as in the first experiment. On top of this pipeline, we introduced an intent classifier and a sentiment classifier to better capture the user's underlying information needs. The intent classifier aims to improve the alignment between user queries and candidate answers by modeling the type of information requested, while the sentiment classifier enables the system to account for the emotional tone of the query.

3.3 Implementation Details

To conduct our study, we employed several tools and frameworks, including PyTerrier, Hugging Face and Doc2Query.

All experiments were based on a standard inverted indexing approach. However, the preprocessing strategies varied depending on the specific experiment, as described below.

Baseline Systems

Experiment 1: Queries were lowercased, with optional retention or removal of non-alphabetical characters.

Experiment 2: Queries were lowercased and cleaned by removing non-alphabetical characters, followed by acronym expansion, morphological expansion, syntactic expansion and query rewriting.

Experiment 3: Longer documents were segmented into shorter passages (chunking). Short and empty documents were filtered out, non-English documents and symbols were discarded, and stopwords were removed to reduce the length of both documents and queries.

Advanced Systems

Experiment 1: Preprocessing included the removal of links and emojis, acronym expansion and document augmentation using Doc2Query.

Experiment 2: The same preprocessing pipeline as in the first advanced experiment was applied.

4 Experiments and Results

4.1 Evaluation Setup

To rigorously assess the performance of our financial retrieval system, we utilized **PyTerrier** as our experimental framework. We evaluated three distinct system configurations: a sparse keyword baseline (BM25), a dense neural re-ranker (E1), and a risk-aware diversity re-ranker (E2). We reported performance using standard metrics, selected to capture different aspects of the user experience: MAP, P@1, P@5, P@10, R@5, R@10, nDCG@5, nDCG@10. All experiments used the same fixed set of train/test splits provided by the dataset. Relevance judgments (qrels) were binary (relevant/non-relevant). To ensure fair comparison, all re-ranking experiments (E1 and E2) operated on the same candidate pool retrieved by the initial BM25 stage.

4.2. Results and Analysis

4.2.1 Best performing baseline

After rigorous testing, we determined that the strongest baseline was BM25 with standard preprocessing (tokenization, lowercase, stopwords removal) on full, un-chunked documents. This "simple" baseline proved surprisingly difficult to beat because it preserved the full semantic context of forum threads while robustly handling length variations via its internal normalization parameters.

Table 1 presents the comparative performance of the three experimental runs.

	name	map	P.1	P.5	P.10	recall.5	recall.10	ndcg_cut_5	ndcg_cut_10
0	BM25	0.207961	0.233025	0.108642	0.072068	0.249016	0.327327	0.230265	0.257055
1	E1	0.285493	0.344136	0.154630	0.093981	0.352451	0.412465	0.328618	0.347721
2	E2	0.279608	0.336420	0.151852	0.093673	0.345472	0.410921	0.321310	0.342616

Table 1 Main Experimental Results

4.2.2 The Semantic Gap (Baseline vs. Experiment 1)

The transition from sparse retrieval (BM25) to neural re-ranking (E1) yielded the most significant performance gains. **Mean Average Precision (MAP)** increased by approximately **37%** (0.208 → 0.285), and **Precision@1** jumped by **48%** (0.233 → 0.344).

This improvement highlights the limitations of keyword matching in the financial domain. Financial queries often suffer from the "vocabulary mismatch" problem (e.g., a user queries "market crash," but relevant documents use technical terms like "correction" or "bearish reversal"). The Cross-Encoder model (E1) successfully bridged this semantic gap, identifying relevant documents that lacked exact keyword overlap but were contextually aligned with the query.

4.2.3 The Relevance-Diversity Trade-off (Experiment 1 vs. Experiment 2)

In Experiment 2, we introduced a "Subjectivity Gate" (threshold >0.6) and a diversity re-ranking logic. This resulted in a slight decrease in traditional relevance metrics compared to E1 (MAP dropped from 0.285 to 0.279; P@1 from 0.344 to 0.336). These metrics do change depending on the threshold set – if it is smaller, then the Diversity Reranker activates more often, thus re-ranks documents more often and so both metrics lower, consequentially, for highest threshold the opposite is true as it comes closer to the “less diverse” E1 system output.

This decrease is expected and represents a deliberate "Diversity Tax." Standard metrics like P@1 and MAP punish any re-ordering that moves a ground-truth "relevant" document down the list. However, our diversity logic intentionally demotes redundant documents (even if relevant) to insert opposing viewpoints.

- **Quantitative Impact:** The drop in P@1 indicates that for some subjective queries, the "most relevant" factual document was displaced by a document expressing a contrarian opinion (e.g., a risk warning).
- **Qualitative Gain:** While metrics slightly degraded, the system gained safety features. For queries classified as "seeking advice," E2 successfully transformed homogenous result lists (e.g., five consecutive "Buy" recommendations) into heterogeneous lists (interleaving "Buy," "Sell," and "Neutral" stances).

4.2.4 Failures and Domain Challenges

A critical failure identified during early iterations was Document Truncation. Initial runs using standard neural scoring yielded poor performance on long conversational threads. Financial advice often appears at the end of long forum discussions; standard models (limited to 512 tokens) ignored this context, scoring relevant documents as irrelevant. We addressed this by implementing a sliding window aggregation strategy for both relevance scoring and sentiment analysis. This ensured that a risk warning buried in the 50th paragraph of a thread was correctly detected and factored into the diversity re-ranking, a step that proved essential for the stability of E2.

5 Discussion and Conclusions

This project highlighted the difficulty of building retrieval systems for highly unstructured and heterogeneous forum data. Data preprocessing emerged as the main challenge, as our analysis revealed substantial differences between queries and documents that complicated both traditional ranking models (e.g., BM25) and neural approaches. As a result, designing preprocessing strategies that aligned with different ranking logics proved non-trivial. Despite these challenges, we focused on developing theory-driven, task-specific retrieval pipelines.

Future work could explore more complex pipelines that combine model outputs using expert judgment and model-specific heuristics. Additional improvements include defining metrics for diversity and developing task-aware notions of “usefulness” and “safety” to better evaluate system performance. Incorporating richer knowledge representations, such as the FIBO ontology, also remains a promising direction.

The system’s primary contribution is its effort to improve access to complex financial information for non-expert users through a balanced retrieval pipeline that explicitly models intent and sentiment. For advice-seeking queries, it reduces potential bias by enforcing sentiment diversity and deliberately surfacing conflicting perspectives, enabling more informed decision-making. This safety objective is supported by a modular pipeline design, with each stage optimized for a specific theoretical role.

AI Tools and Assistance Declaration

- ChatGPT - generation of the expansion of the acronyms, brainstorming on possible usefulness of discussed pipelines, aid in paraphrasing passages of the report to shorten them
- Gemini - Used to explain some code, aid in bug fixing